



From Data to Solutions · Weekend 7

Distributed gradient descent: splitting the batch

Part 1 of 2: when the model fits and the data does not

Carlos Cotrini



Section 1

The gradient of a mean

1

Paris, 1793: one task, twenty computers

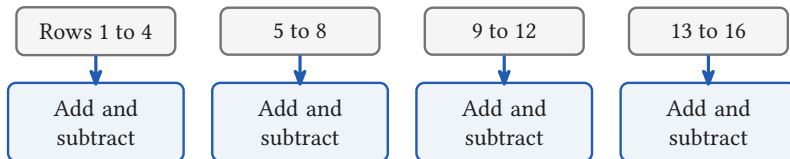
Rows 1 to 4

5 to 8

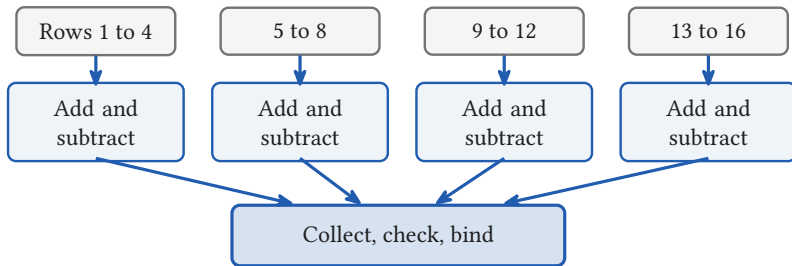
9 to 12

13 to 16

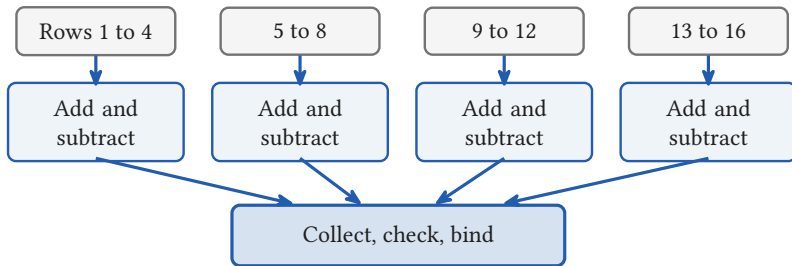
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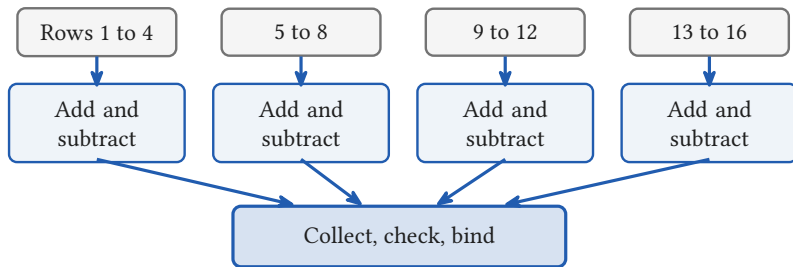


Paris, 1793: one task, twenty computers



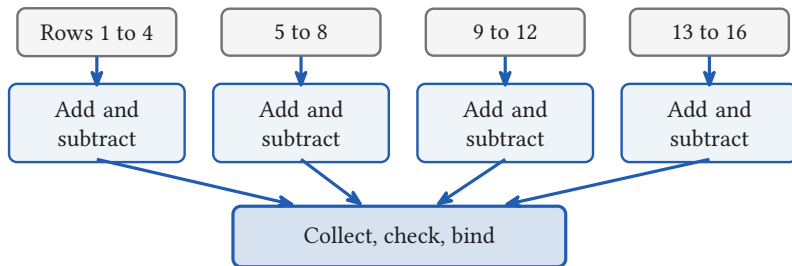
- The same arithmetic in every hand, a different slice of the table

Paris, 1793: one task, twenty computers



- The same arithmetic in every hand, a different slice of the table
- Not a pin factory: nobody specialises in a step

Paris, 1793: one task, twenty computers



- The same arithmetic in every hand, a different slice of the table
- Not a pin factory: nobody specialises in a step
- [Roegel 2010](#), from the payrolls: about twenty computers, not the sixty to eighty Prony recalled.

The sum splits, exactly

$$L(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n \ell_i(\mathbf{w})$$

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$$\nabla L(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n \nabla \ell_i(\mathbf{w})$$

The sum splits, exactly

$$\nabla L(\mathbf{w}) = \frac{1}{P} \sum_{p=1}^P \underbrace{\frac{P}{n} \sum_{i \in S_p} \nabla \ell_i(\mathbf{w})}_{\text{worker } p \text{ computes this alone}}$$

The batch, n examples



P workers, n/P examples each

The sum splits, exactly

$$\nabla L(w) = \frac{1}{P} \underbrace{\sum_{p=1}^P g_p}$$

the only thing the workers must share

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- Equal shards, so every worker holds the same number of examples

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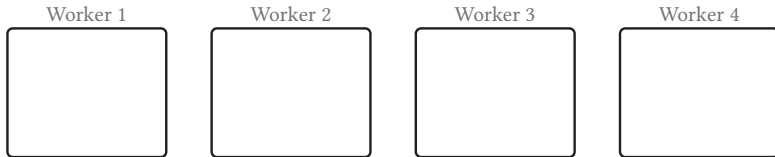
The batch, n examples



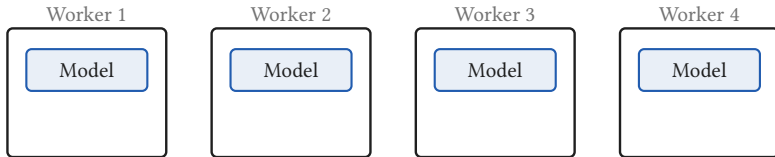
P workers, n/P examples each

- Equal shards, so every worker holds the same number of examples
- You already have this from Weekend 0. Nothing here is new mathematics.

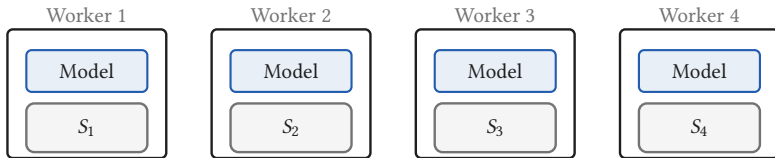
Every worker holds the whole model



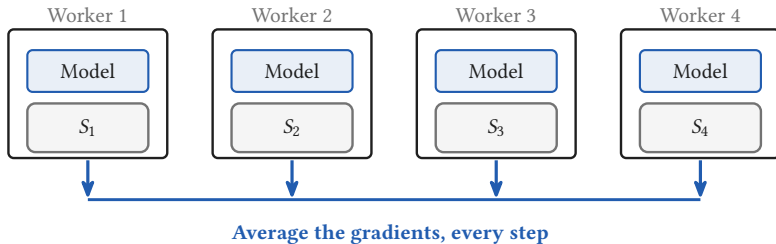
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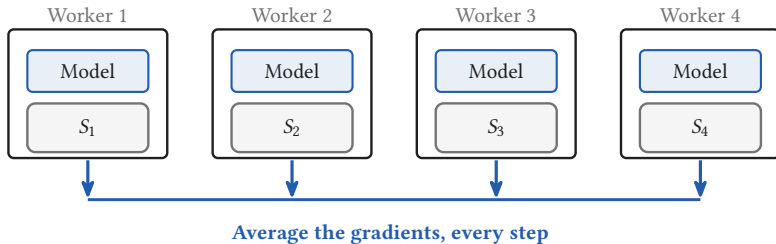
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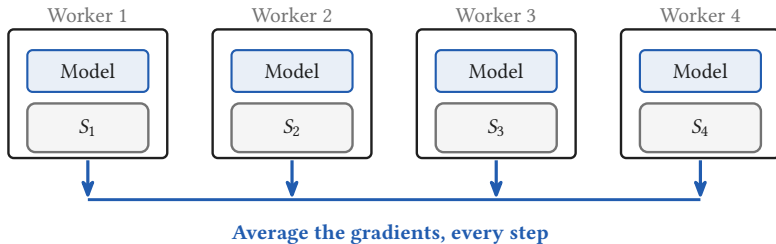


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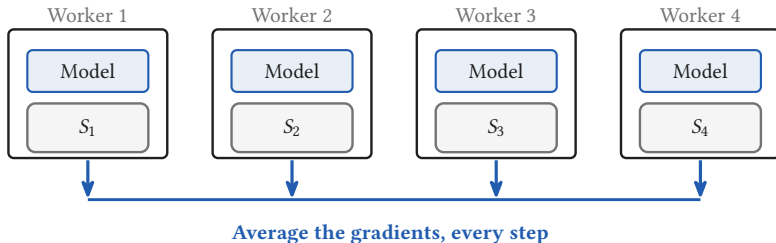
- Data parallelism: replicate the model, split the batch

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- The copies stay identical because every worker applies the same average

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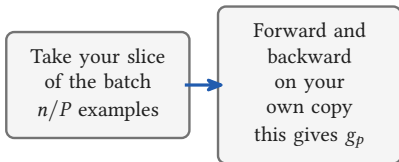


- Data parallelism: replicate the model, split the batch
- The copies stay identical because every worker applies the same average
- This hour assumes the model fits on one device

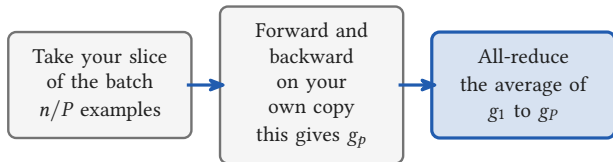
One step of training, run on every worker at once

Take your slice
of the batch
 n/P examples

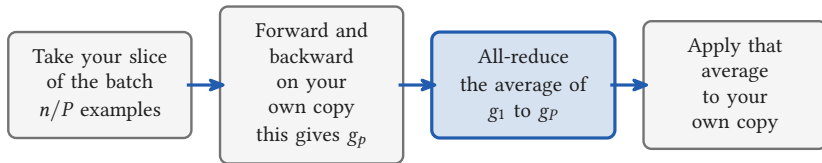
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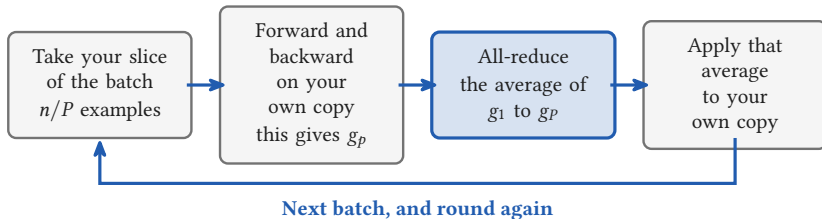
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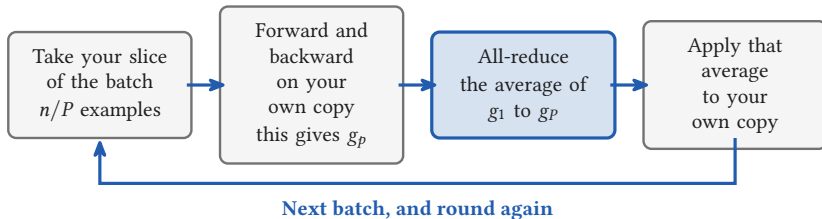
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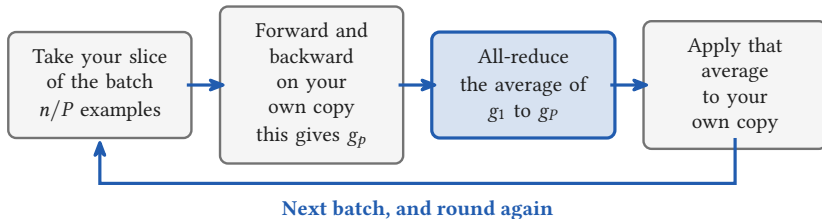


One step of training, run on every worker at once



Three of the four stages are local. The all-reduce is the one that touches the network.

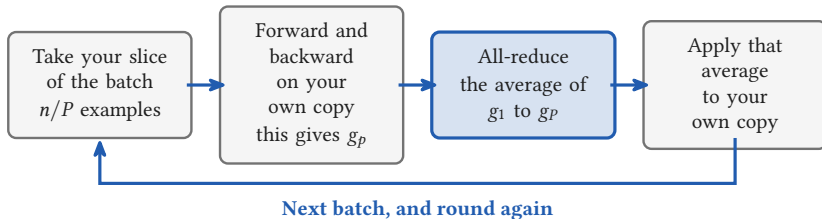
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One step of training, run on every worker at once



Three of the four stages are local. The all-reduce is the one that touches the network.

- Every worker runs this same loop, at the same time, on its own slice
- Split the data, aggregate the gradient, and repeat

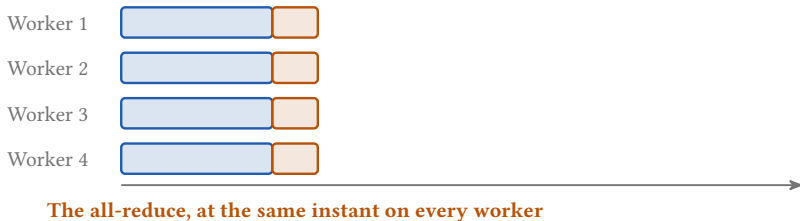
The gradient of a mean

One all-reduce per step, for every step of the run



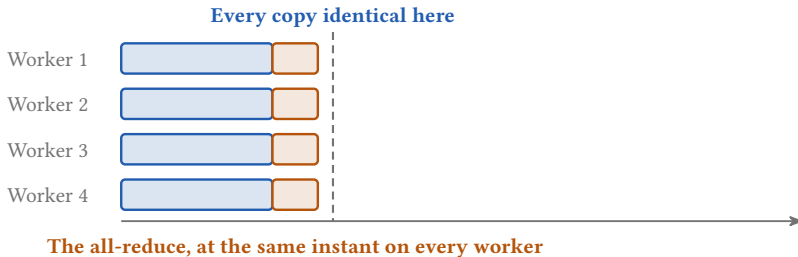
☰ The gradient of a mean

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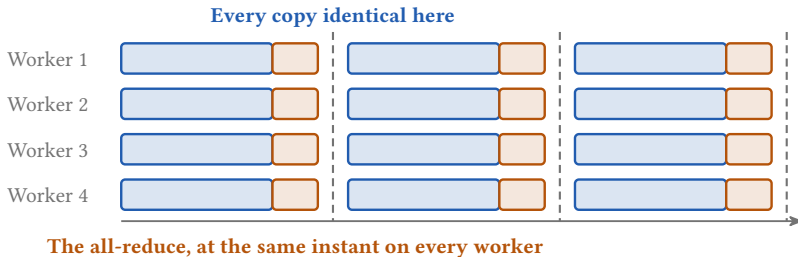


The gradient of a mean

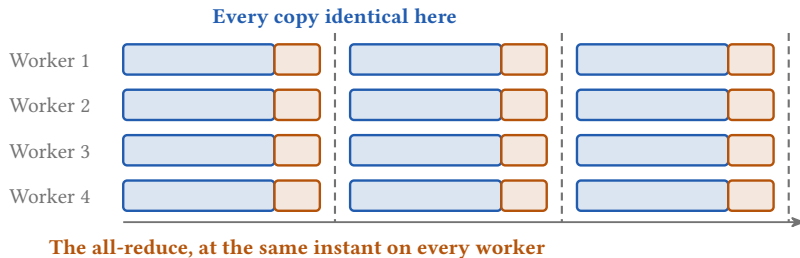
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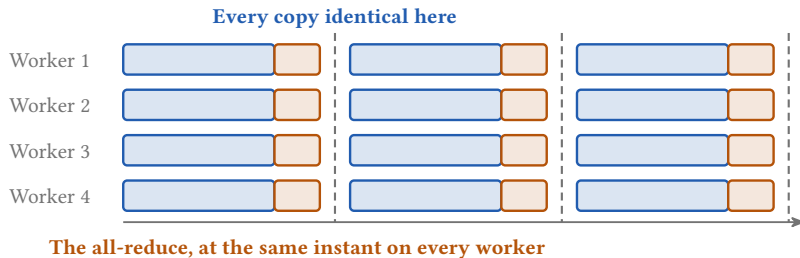


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One all-reduce per step, for every step of the run



- The average lands before the weights move, so every step starts from one model
- Local SGD relaxes this and aggregates every few steps. Everything in this hour is the synchronous version.



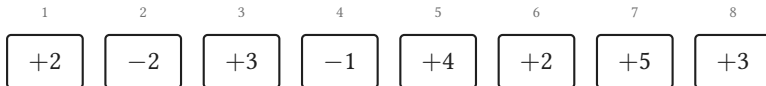
Section 2

Four workers, eight examples

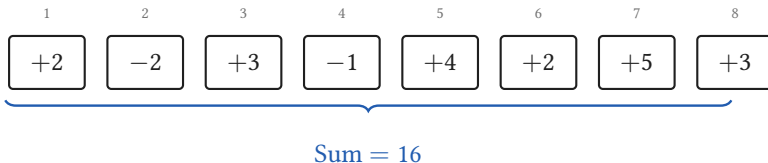
2

Four workers, eight examples

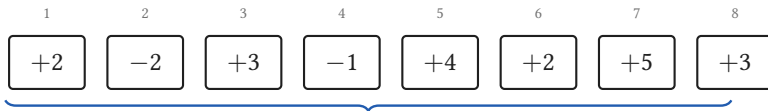
Eight examples, one gradient



Eight examples, one gradient



Eight examples, one gradient

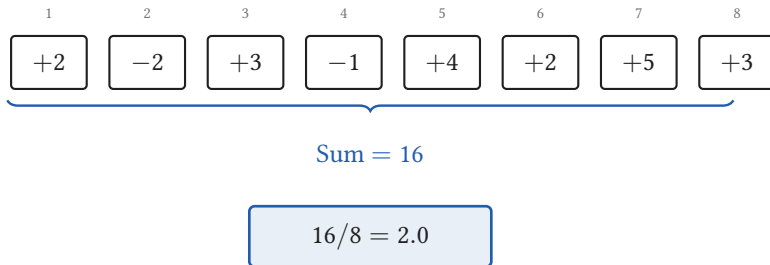


Sum = 16

$$16/8 = 2.0$$

- The single machine answer is 2.0

Eight examples, one gradient

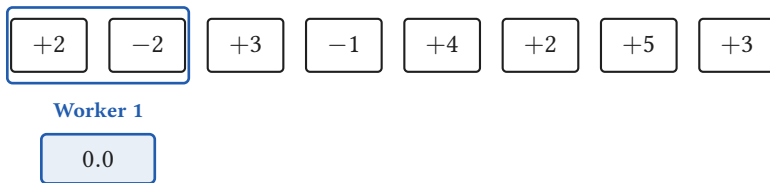


- The single machine answer is 2.0
- Constructed for this lecture. A real gradient is a vector with one number per parameter; one number per example keeps the arithmetic hand checkable.

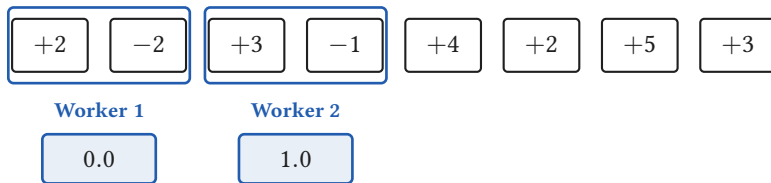
Four workers, two examples each



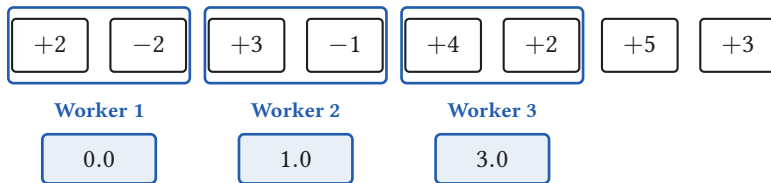
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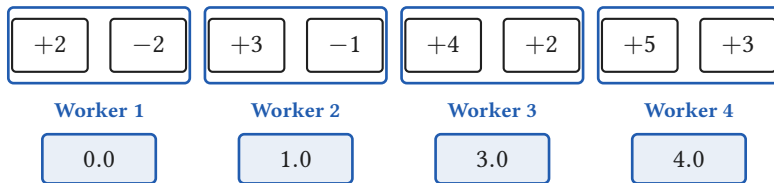
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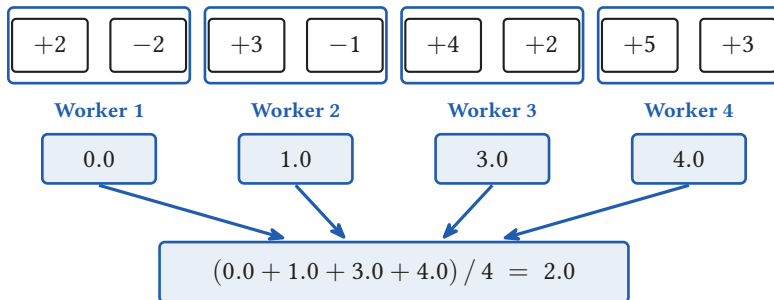
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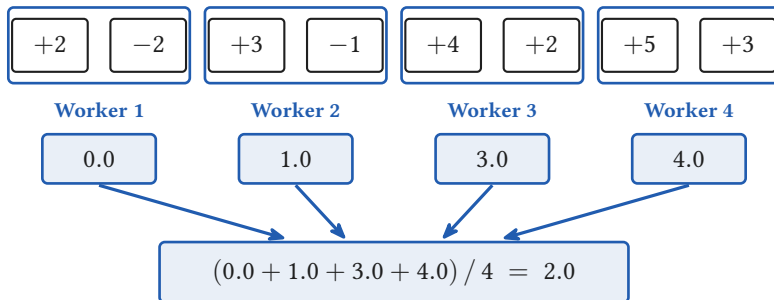
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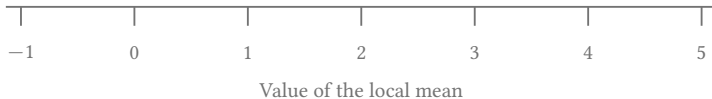


The same number the single machine got

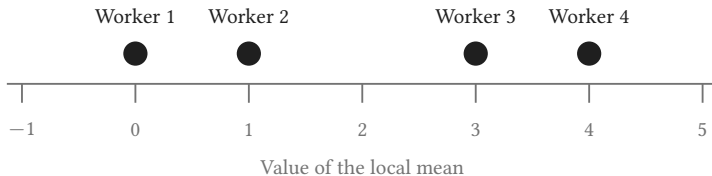
Exactly equal, not approximately: splitting the batch is a regrouping of the same sum.

Four workers, eight examples

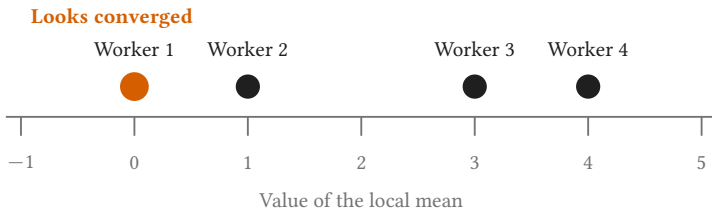
Worker 1 sees zero



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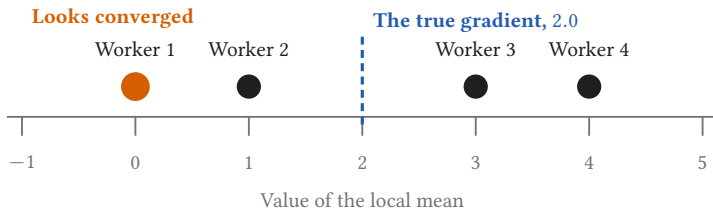


Worker 1 sees zero



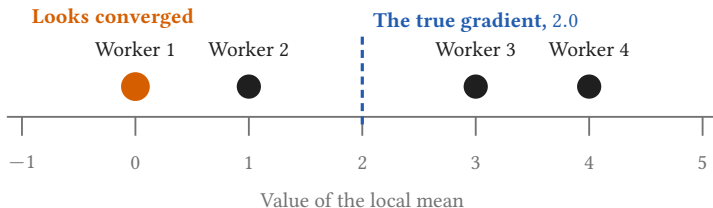
- Worker 1's local mean is zero. Its own shard looks converged.

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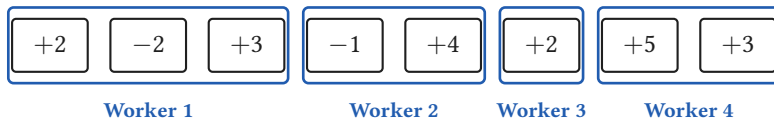
- Worker 1's local mean is zero. Its own shard looks converged.
- The global answer is 2.0, and no worker holds it

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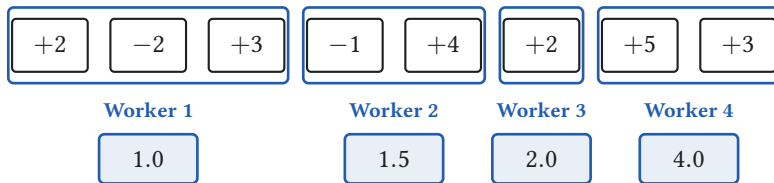


- Worker 1's local mean is zero. Its own shard looks converged.
- The global answer is 2.0, and no worker holds it
- That is why the averaging step exists, and why it cannot be skipped

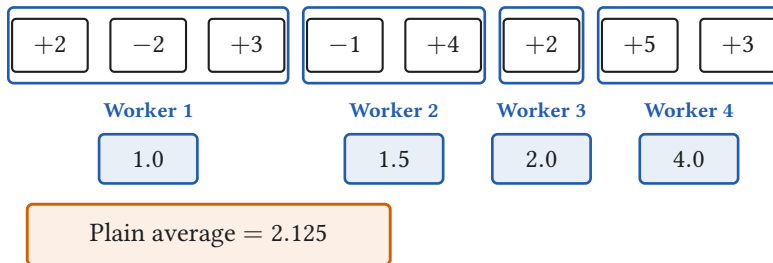
Unequal shards break the identity



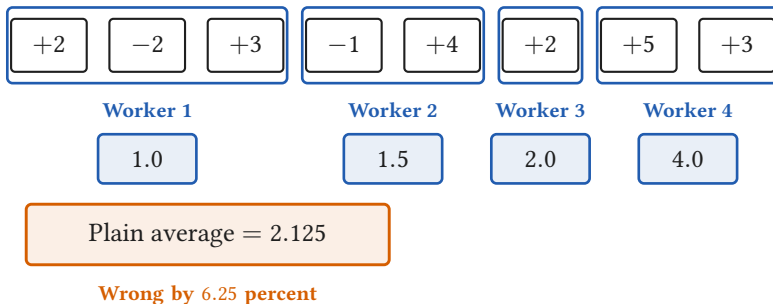
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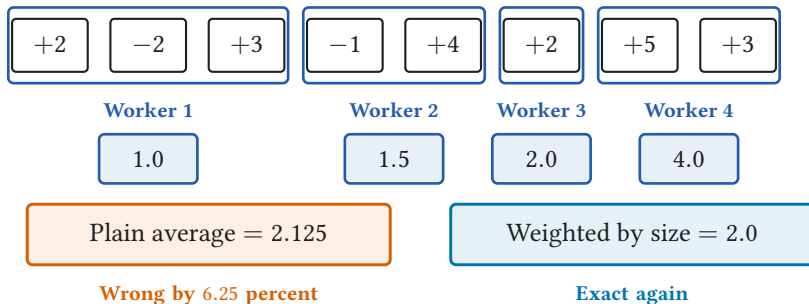
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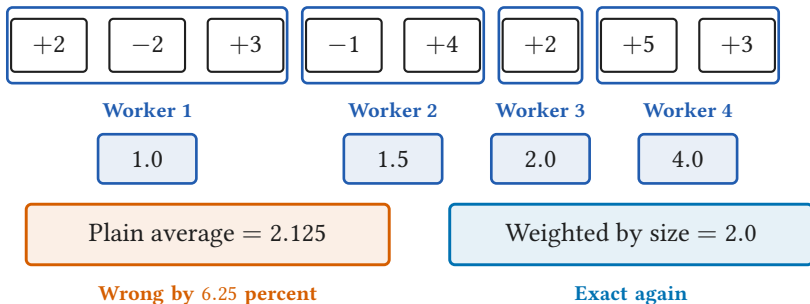
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Unequal shards break the identity



Every framework therefore fixes the per worker micro batch and drops or pads the ragged tail.

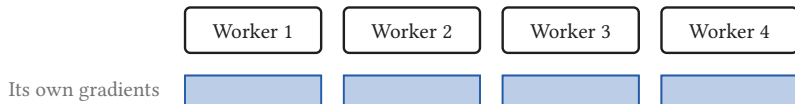
Four workers, eight examples

One buffer, 16.11 gigabytes



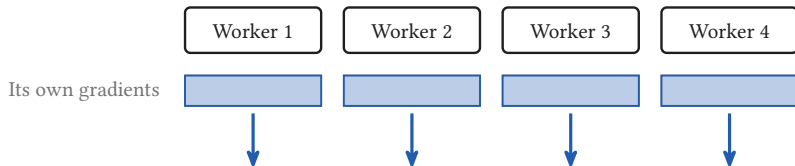
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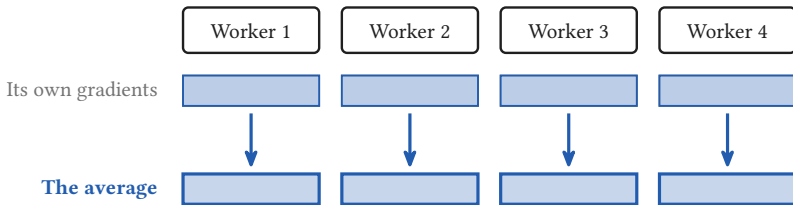


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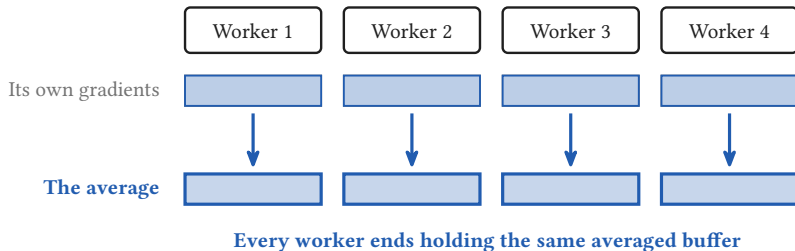


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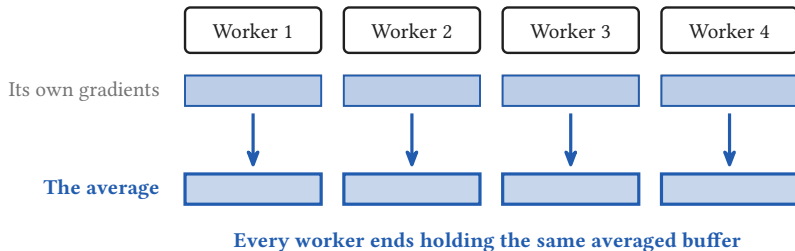
Every worker ends holding the same averaged buffer

One buffer, 16.11 gigabytes



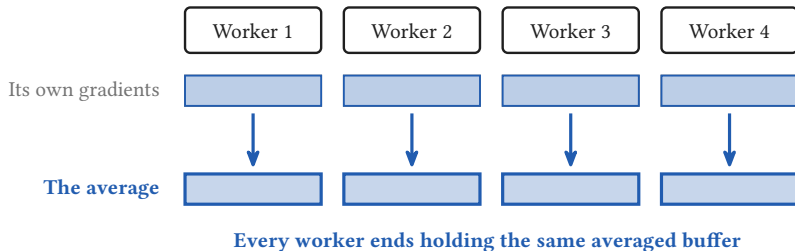
- Apertus-8B in bf16: $2 \text{ bytes} \times 8.053 \times 10^9 \text{ parameters} = 16.11 \text{ GB}$

One buffer, 16.11 gigabytes



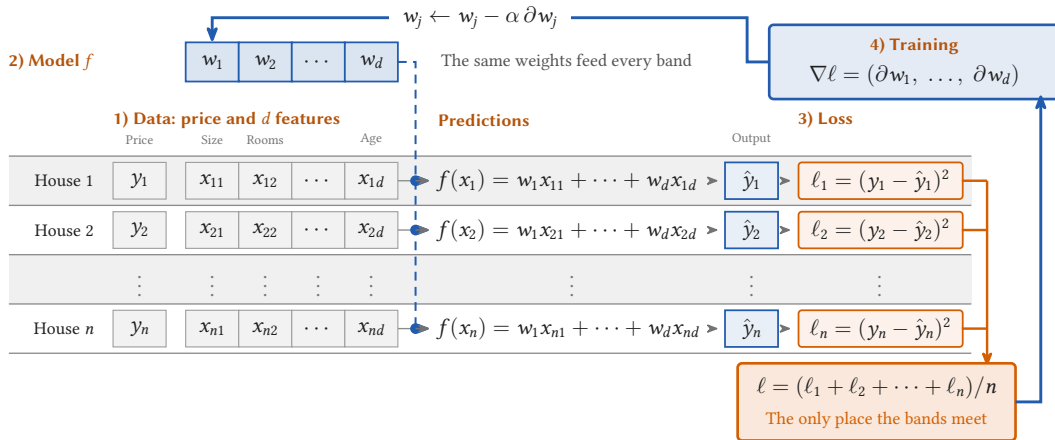
- Apertus-8B in bf16: $2 \text{ bytes} \times 8.053 \times 10^9 \text{ parameters} = 16.11 \text{ GB}$
- Write K for the size of that buffer, and P for the number of workers

One buffer, 16.11 gigabytes



- Apertus-8B in bf16: $2 \text{ bytes} \times 8.053 \times 10^9 \text{ parameters} = 16.11 \text{ GB}$
- Write K for the size of that buffer, and P for the number of workers
- Compute is local. This is the only part that is not.

One training step of linear regression



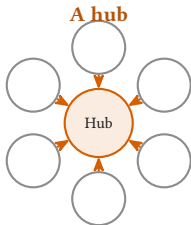


Section 3

The all-reduce

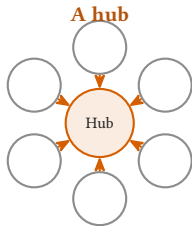
3

Two ways to average: a hub, or a ring



K is the gradient buffer, 16.11 GB. P is the number of workers.

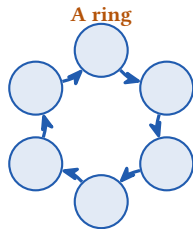
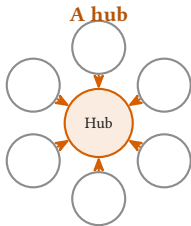
Two ways to average: a hub, or a ring



$P \times K$ arrives at one machine

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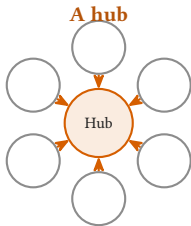
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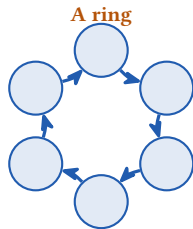
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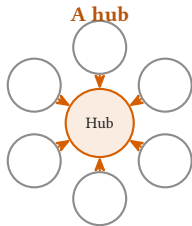
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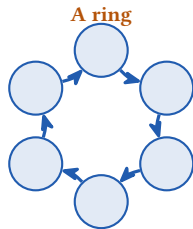
$\frac{2(P-1)}{P} \times K$ leaves each worker

Two ways to average: a hub, or a ring



$P \times K$ arrives at one machine

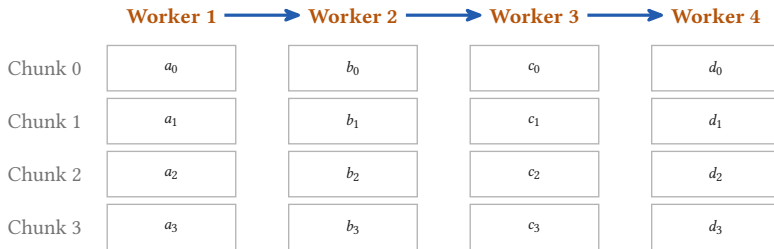
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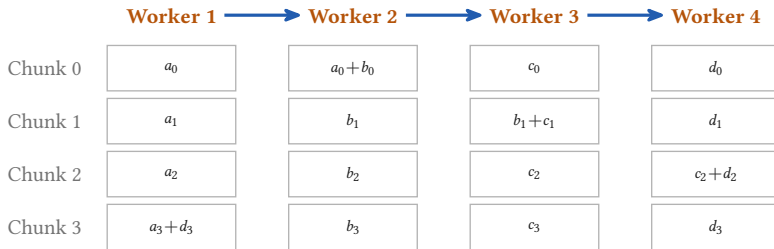
- The hub's traffic grows with every worker you add

Round the ring, adding



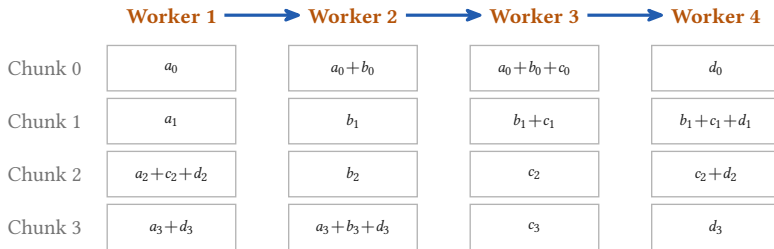
Start: every worker holds its own four chunks

Round the ring, adding



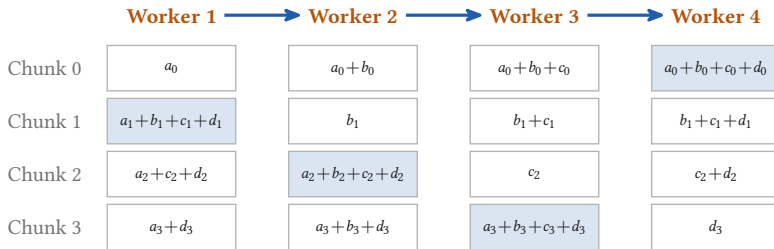
Round 1: send one chunk to the right, add what arrives

Round the ring, adding



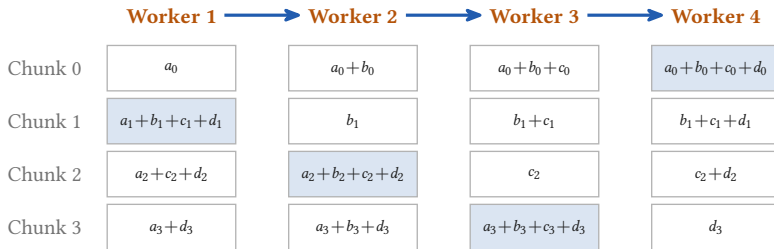
Round 2: send on the chunk you have just added to

Round the ring, adding



Round 3: each worker now owns one chunk of the true sum

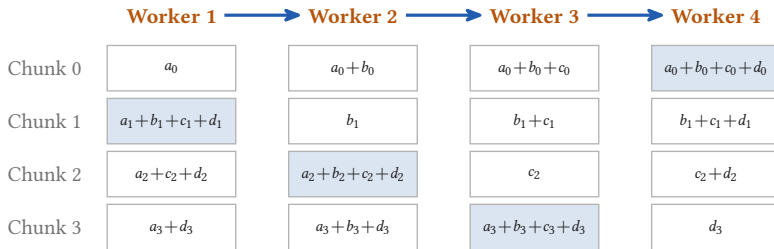
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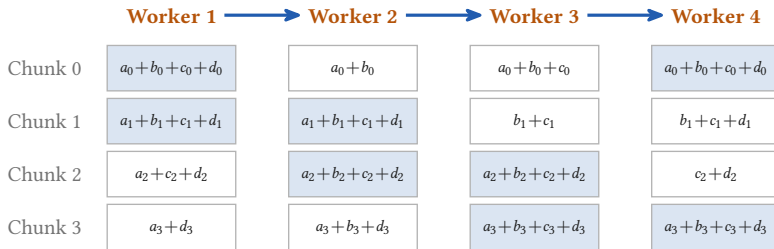
Chunk j is the same slice of the buffer on every worker, so a_j is worker 1's chunk j , b_j worker 2's, and worker 4 sends back to worker 1.

Round the ring again, copying



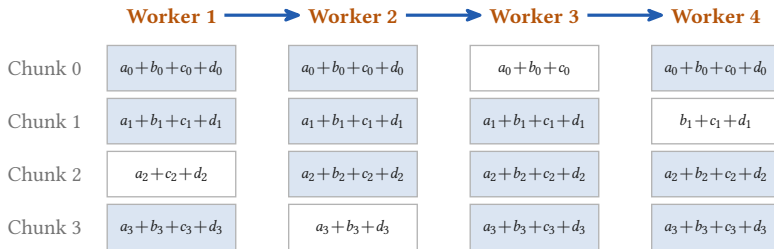
Where the first half ended: one complete chunk each

Round the ring again, copying



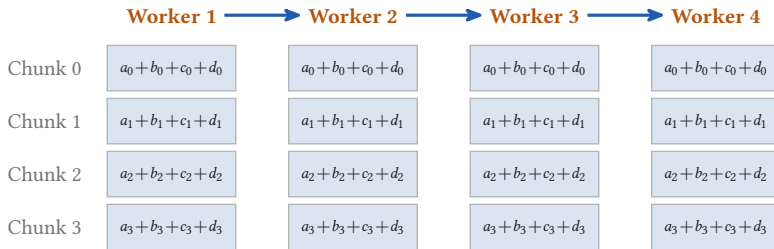
Round 4: send the complete chunk on, and overwrite

Round the ring again, copying



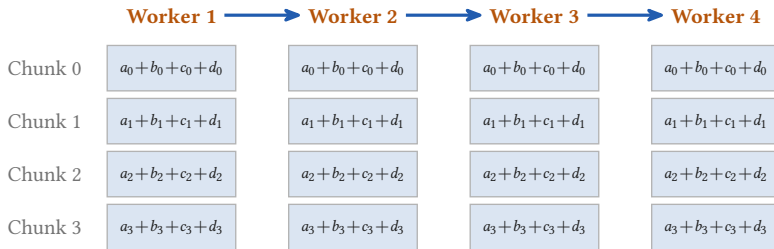
Round 5: the same traffic, one hop further

Round the ring again, copying



Round 6: every worker now holds the whole averaged buffer

Round the ring again, copying



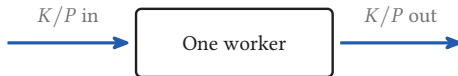
Round 6: every worker now holds the whole averaged buffer

Every worker holds the whole model, so the copies stay identical only if every worker applies the same whole average

Same ring, same traffic, one change: the arriving chunk replaces instead of adding. The four rows are four different slices, so a column of four cells is the whole averaged buffer.

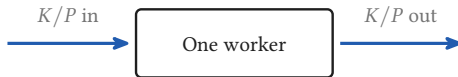
Two times P minus one rounds

$$\text{bytes per worker} = 2(P - 1) \times \frac{K}{P}$$



Two times P minus one rounds

$$\text{bytes per worker} = \underbrace{2(P-1)}_{\text{rounds}} \times \underbrace{\frac{K}{P}}_{\text{one chunk}}$$

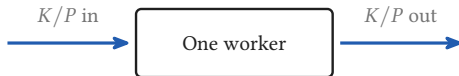


$P - 1$ rounds adding, then $P - 1$ copying

A chunk has to reach the other $P - 1$ workers, so one pass round the ring is $P - 1$ rounds, three of them at $P = 4$

Two times P minus one rounds

$$\text{bytes per worker} = \underbrace{\frac{2(P-1)}{P}}_{\text{always below 2}} \times \underbrace{K}_{\text{the whole buffer}}$$

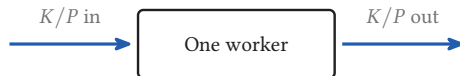


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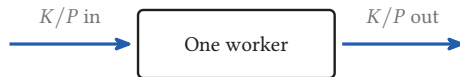
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A chunk has to reach the other $P - 1$ workers, so one pass round the ring is $P - 1$ rounds, three of them at $P = 4$

- The factor never reaches 2, so the traffic never reaches $2K$
- [Patarasuk and Yuan 2009](#) proved this is the least any all-reduce can move.

The bytes barely grow

$P = 2$

16.11 GB

The bytes barely grow

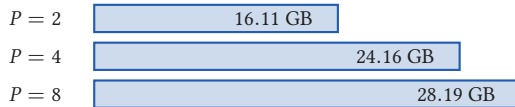
$P = 2$

16.11 GB

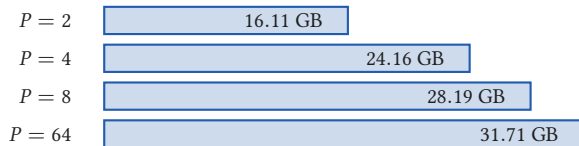
$P = 4$

24.16 GB

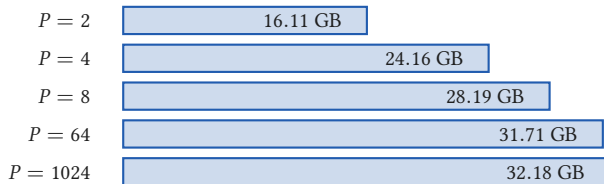
The bytes barely grow



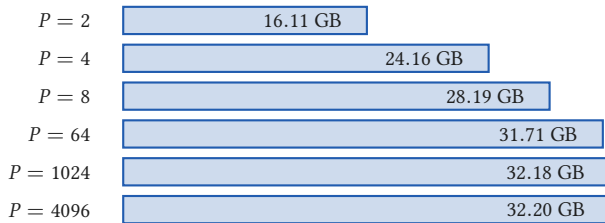
The bytes barely grow



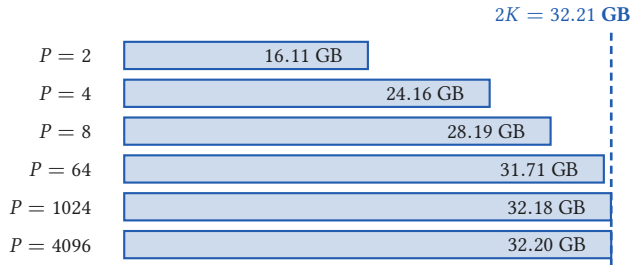
The bytes barely grow



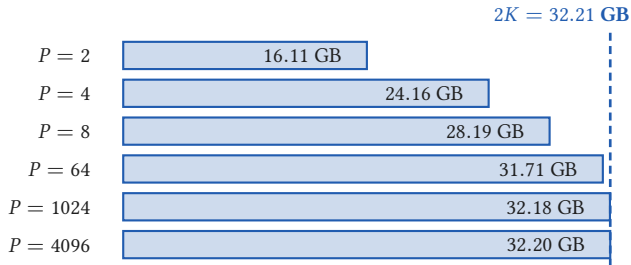
The bytes barely grow



The bytes barely grow

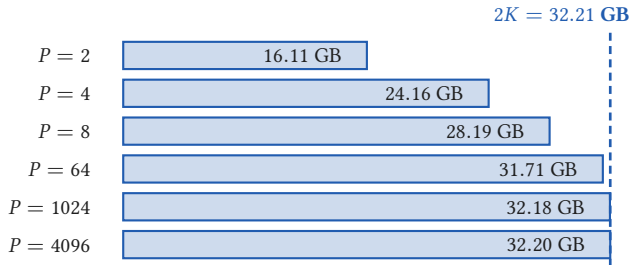


The bytes barely grow



- 512 times the workers, 14.3 percent more bytes each

The bytes barely grow



- 512 times the workers, 14.3 percent more bytes each
- Apertus-8B gradients in bf16, so $K = 16.11$ GB.

The rounds grow linearly in P

$P = 8$ | 14 rounds

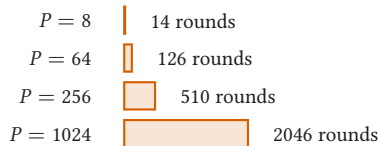
The rounds grow linearly in P

$P = 8$		14 rounds
$P = 64$		126 rounds

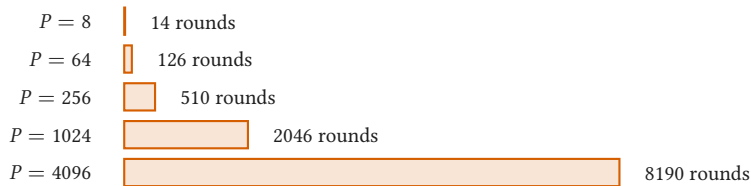
The rounds grow linearly in P

$P = 8$		14 rounds
$P = 64$		126 rounds
$P = 256$		510 rounds

The rounds grow linearly in P

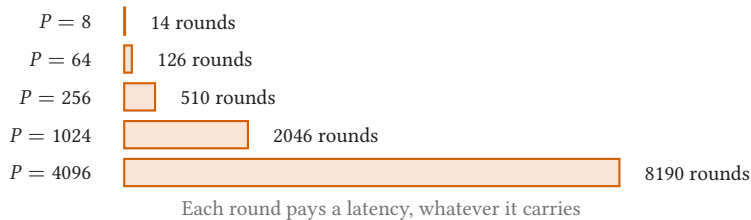


The rounds grow linearly in P



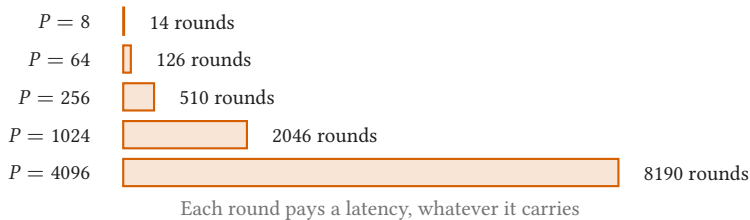
Each round pays a latency, whatever it carries

The rounds grow linearly in P



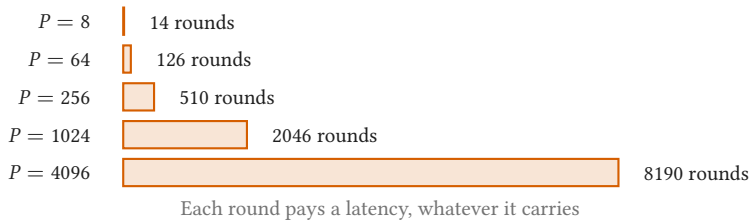
- The bytes barely grow. The rounds grow linearly.

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The rounds grow linearly in P



- The bytes barely grow. The rounds grow linearly.
- Past 512 GPUs the latency no longer hides behind the backward pass
- The ring is not Baidu's invention: it is [Patarasuk and Yuan 2009](#).



Section 4

Where it stops scaling

4

One step, two terms

$T(P)$ is the wall clock time for *one training step* on P workers

$$T(P) = \frac{T(1)}{P}$$

One step, two terms

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$$T(P) = \underbrace{\frac{T(1)}{P}}_{\text{falls with } P} + \text{communication}$$

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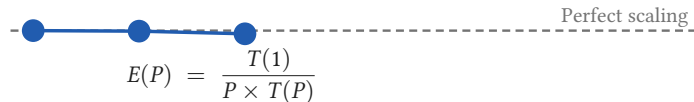
The buffer K : 16.11 GB, the gradients of Apertus-8B in bf16

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$$T(1024) = \underbrace{1128/1024}_{\text{compute}} + \underbrace{1.29}_{\text{all-reduce}} = 2.39 \text{ seconds}$$

Where it stops scaling

93 percent at 64 workers, 18 at 4,096

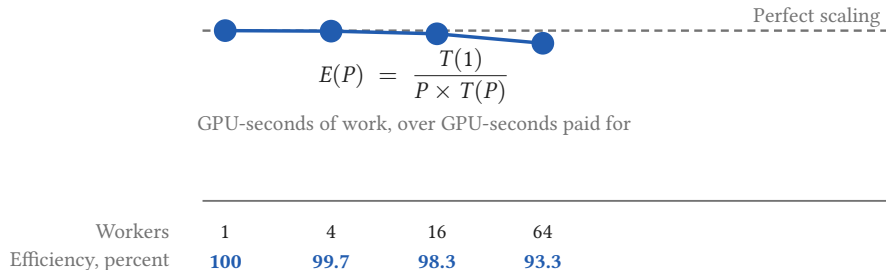


GPU-seconds of work, over GPU-seconds paid for

Workers	1	4	16
Efficiency, percent	100	99.7	98.3

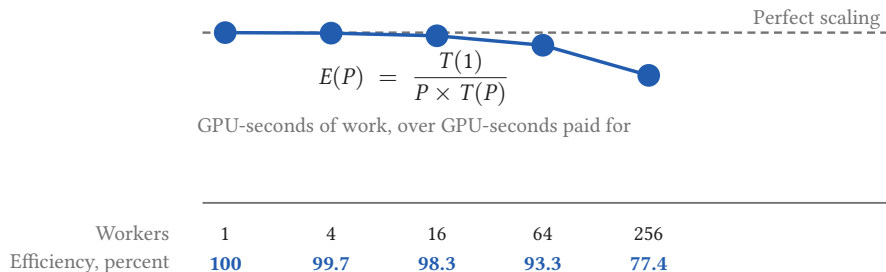
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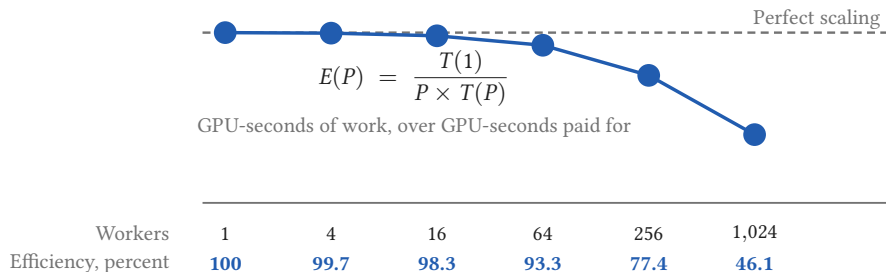
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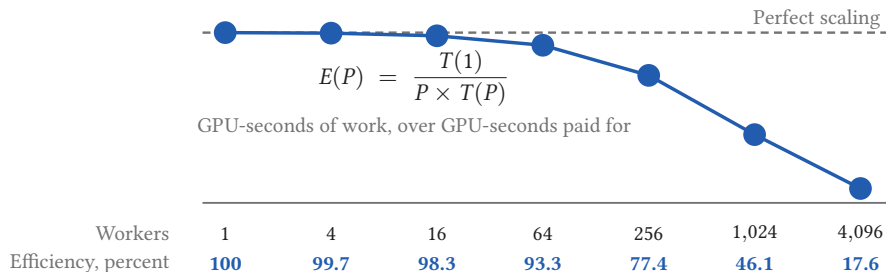
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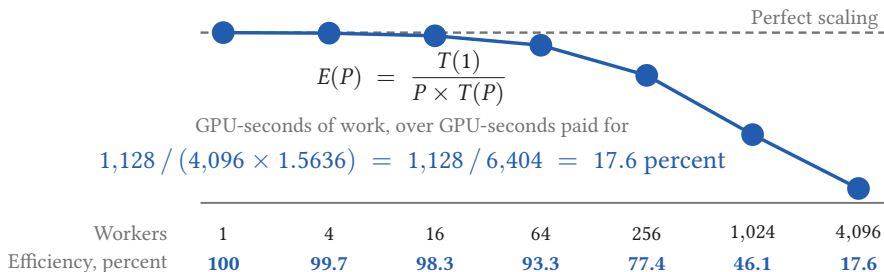
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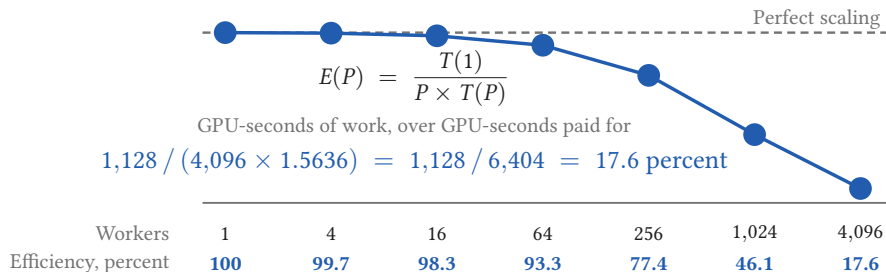
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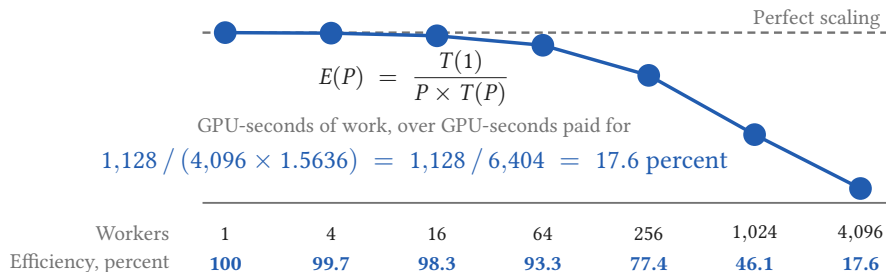
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- So 82 percent of the cluster-seconds in that step were spent waiting

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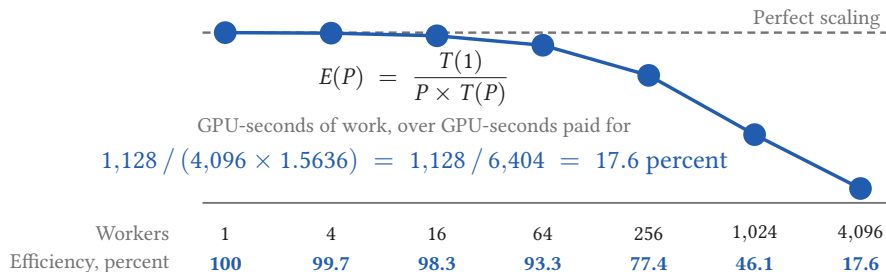
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- At 1,024 workers, communication is 54 percent of the step

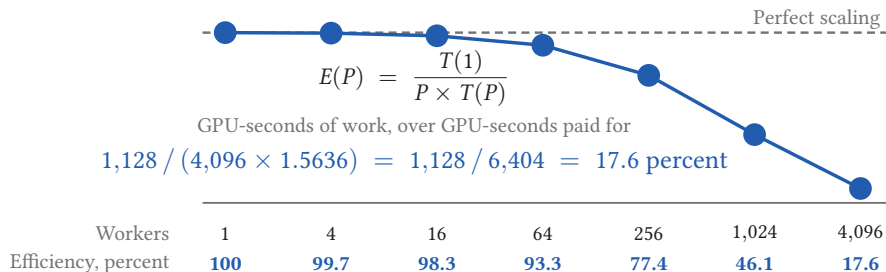
Where it stops scaling

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- The first term falls as one over P . The second does not fall at all.

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- So 82 percent of the cluster-seconds in that step were spent waiting
- At 1,024 workers, communication is 54 percent of the step
- The first term falls as one over P . The second does not fall at all.
- These are the model's numbers, charging the all-reduce after the whole backward pass.

Where it stops scaling

In this model the step never falls below 1.29 seconds

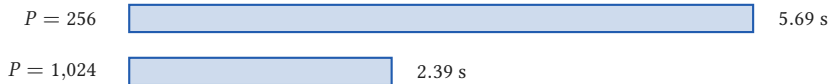
$P = 256$



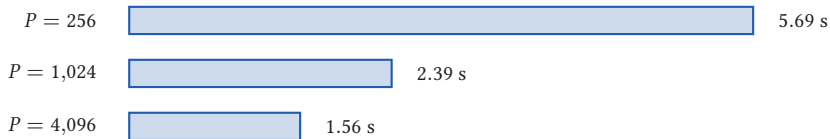
5.69 s

Where it stops scaling

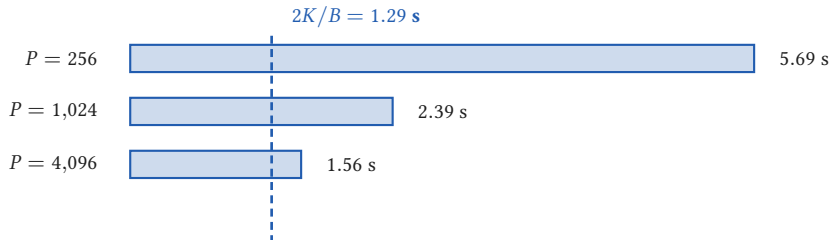
In this model the step never falls below 1.29 seconds



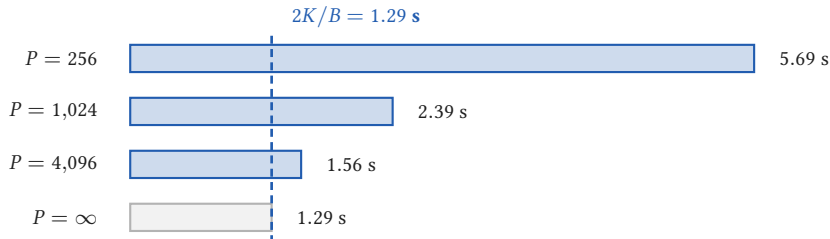
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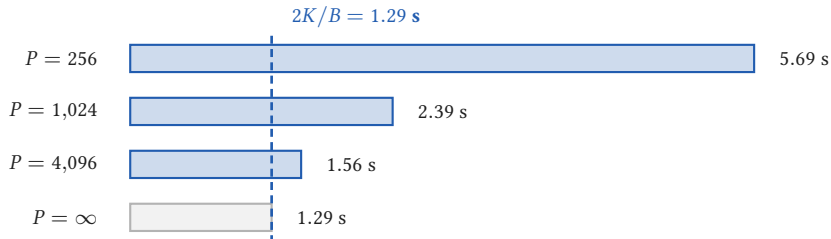
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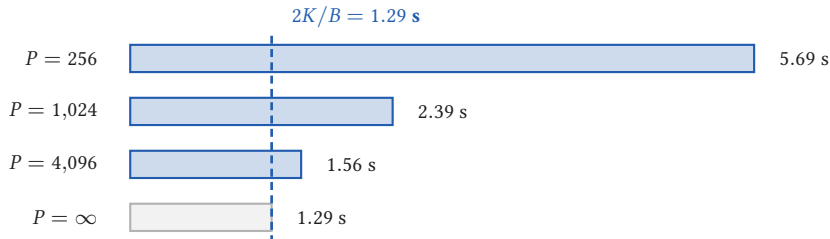


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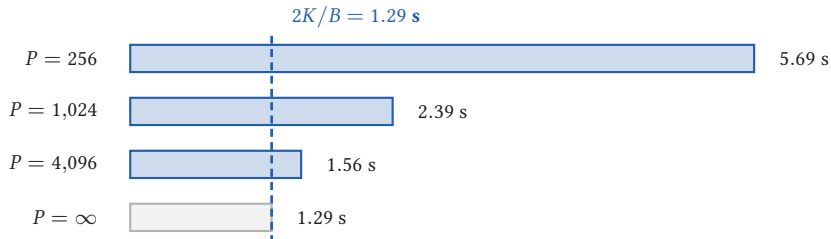
- Compute over P goes to zero, and $2(P - 1)/P \times K/B$ goes to $2K/B$

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- Four times the machine, 1,024 to 4,096, buys 0.83 s of a 2.39 s step

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- Compute over P goes to zero, and $2(P - 1)/P \times K/B$ goes to $2K/B$
- Four times the machine, 1,024 to 4,096, buys 0.83 s of a 2.39 s step
- Every worker after that, all of them together, buys 0.28 s more

Where it stops scaling

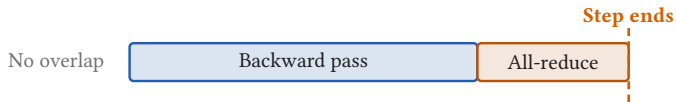
18 percent modelled, 80 percent measured

No overlap

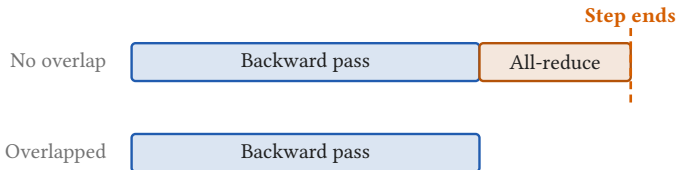
Backward pass

Where it stops scaling

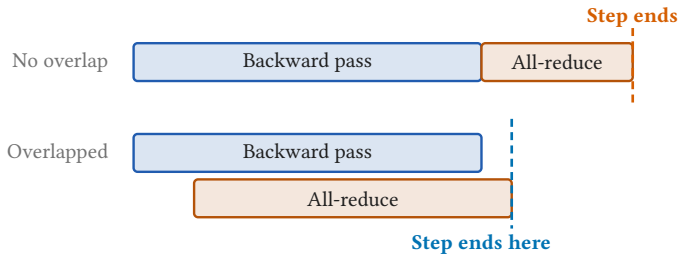
18 percent modelled, 80 percent measured



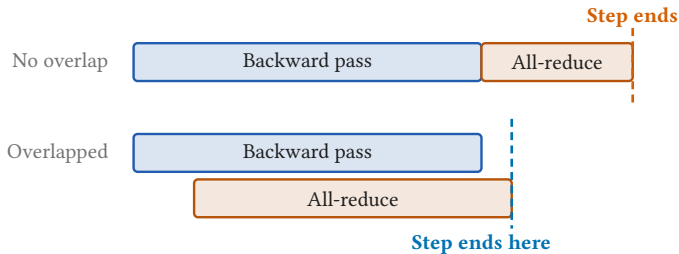
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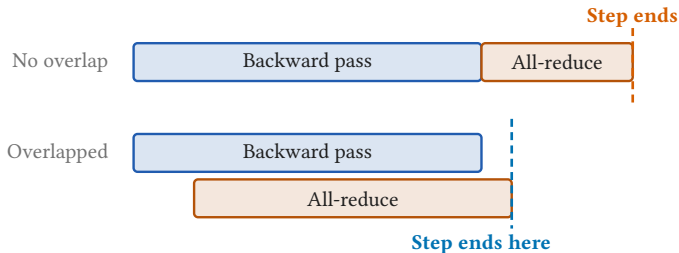


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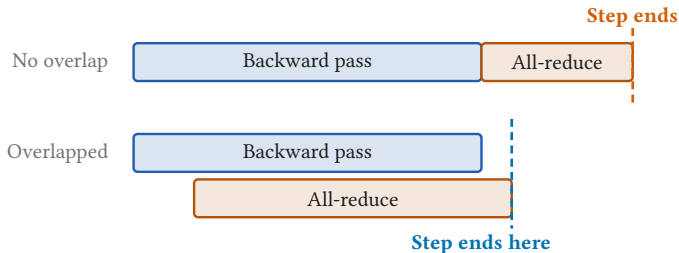
- The model above charges the all-reduce after the whole backward pass

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- Real systems start averaging before the backward pass has finished

18 percent modelled, 80 percent measured



- The model above charges the all-reduce after the whole backward pass
- Real systems start averaging before the backward pass has finished
- Apertus measured about 80 percent at 4,096 GPUs, not 17.6, so overlapping is worth a factor of about 4.5

Where it stops scaling

29 hours on 8 GPUs, 1 hour on 256

8 GPUs, batch 256



29 hours

Where it stops scaling

29 hours on 8 GPUs, 1 hour on 256



Where it stops scaling

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Ideal 32 times faster, measured 29 times: 90.6 percent

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- Accuracy is stable to about 8k images and degrades rapidly beyond

Where it stops scaling

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Ideal 32 times faster, measured 29 times: 90.6 percent

- Time per iteration rises only 12 percent while the minibatch grows 44 times
- Accuracy is stable to about 8k images and degrades rapidly beyond
- [Goyal et al. 2017](#), ResNet-50 on ImageNet.

Multiply the batch by k, the rate by k

Baseline, batch 256

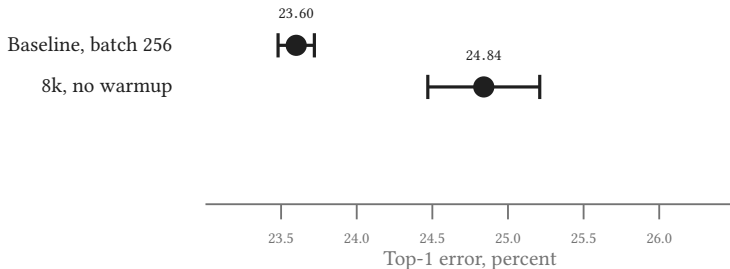
23.60



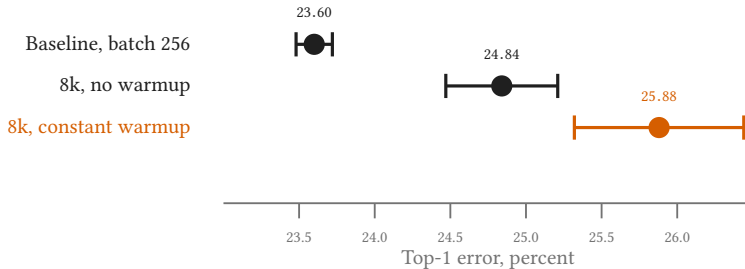
23.5 24.0 24.5 25.0 25.5 26.0

Top-1 error, percent

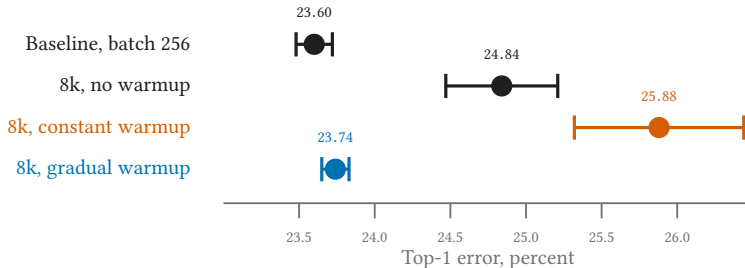
Multiply the batch by k, the rate by k



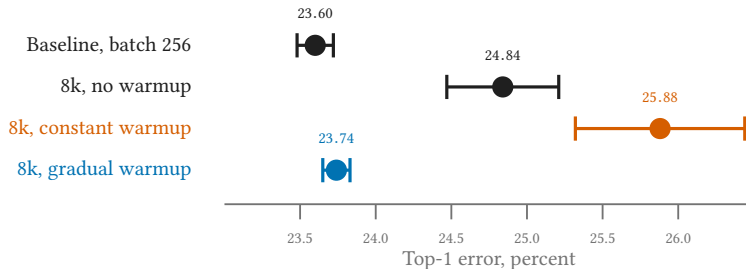
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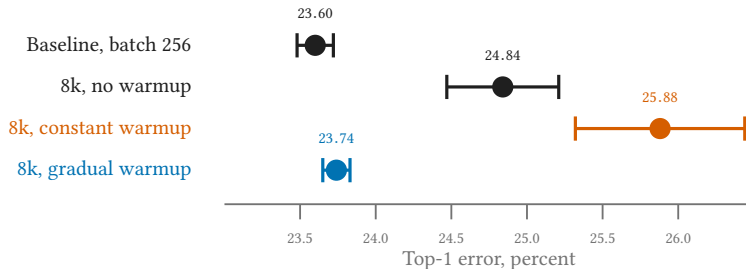


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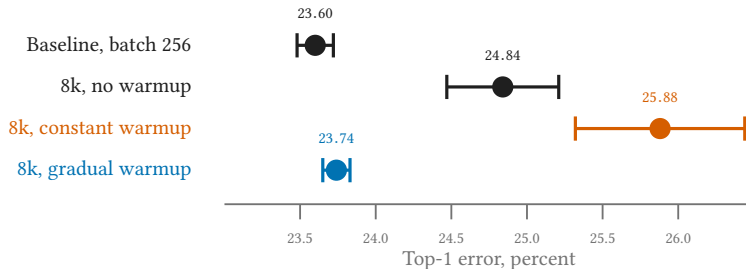
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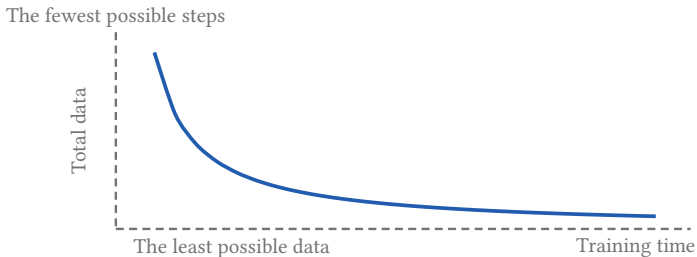


- Constant warmup is worse than no warmup at all
- Gradual warmup over five epochs recovers the baseline
- [Goyal et al. 2017](#) Table 1, five trials each. An SGD result: do not carry the linear rule to an Adam run.

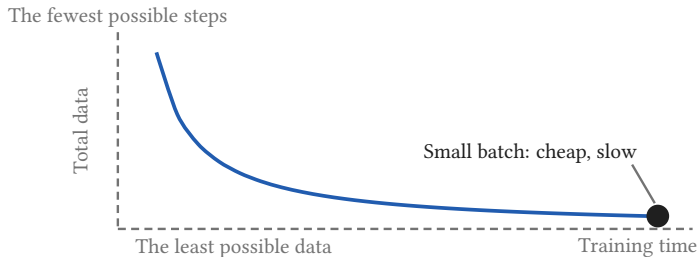
Past the critical batch size



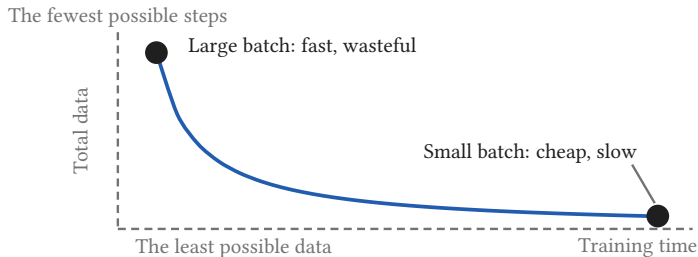
Past the critical batch size



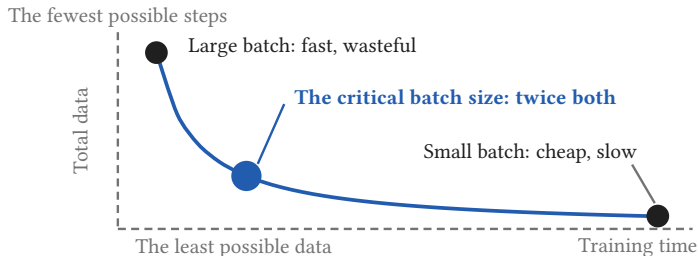
Past the critical batch size



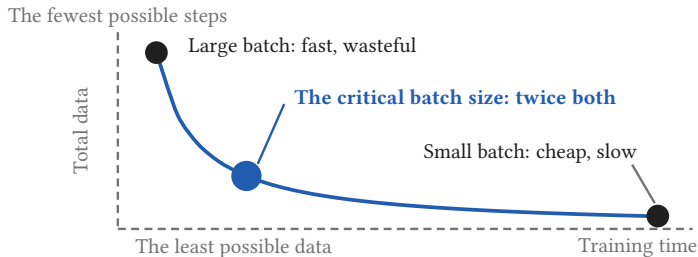
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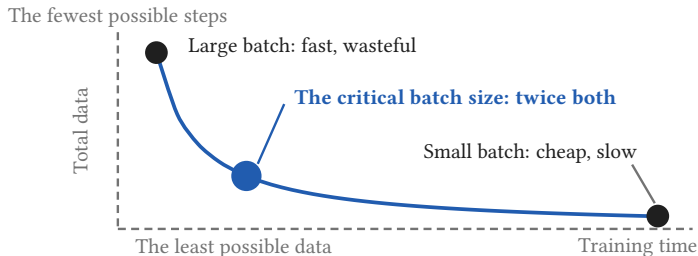


Past the critical batch size



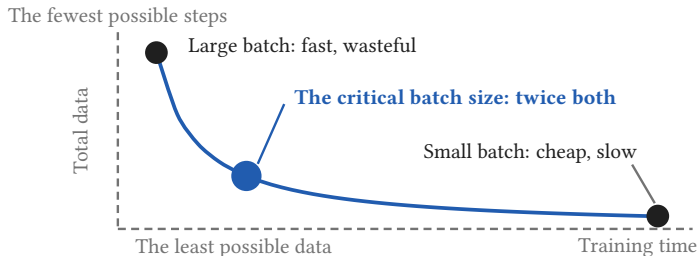
- Twice the data of the cheapest run, twice the steps of the fastest

Past the critical batch size



- Twice the data of the cheapest run, twice the steps of the fastest
- Measured, not derived: Billion Word 700 to 100,000 tokens

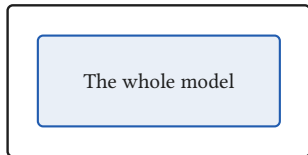
Past the critical batch size



- Twice the data of the cheapest run, twice the steps of the fastest
- Measured, not derived: Billion Word 700 to 100,000 tokens
- It rises during a run, so Apertus doubled its batch mid run

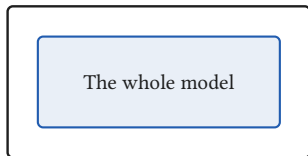
Tomorrow the model does not fit

Today



Tomorrow the model does not fit

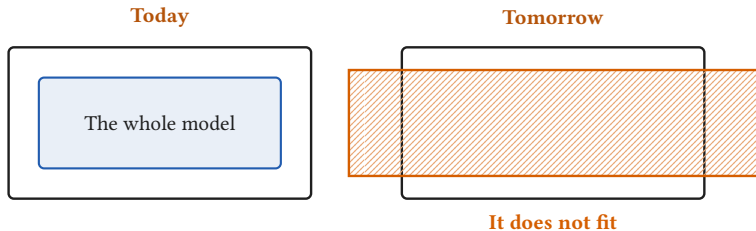
Today



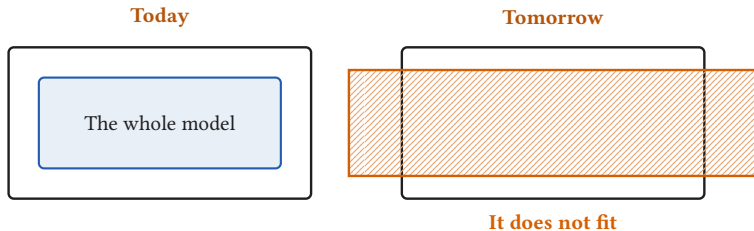
Tomorrow



Tomorrow the model does not fit

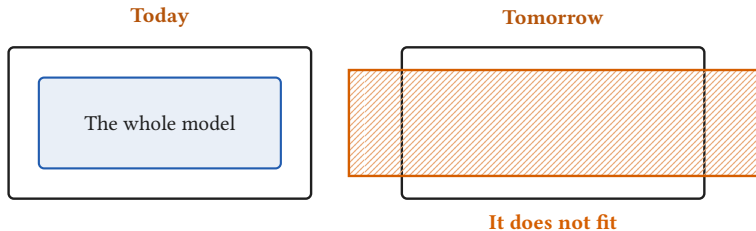


Tomorrow the model does not fit



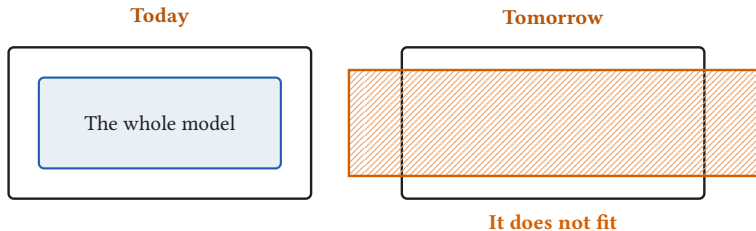
- Everything in this hour assumed one device holds the whole model

Tomorrow the model does not fit



- Everything in this hour assumed one device holds the whole model
- Apertus-70B does not

Tomorrow the model does not fit



- Everything in this hour assumed one device holds the whole model
- Apertus-70B does not
- Tomorrow at 08:00: cut the model, not the batch